The quadratic formula and the discriminant

One formula that solves every quadratic — and one number inside it that says in advance how many roots there will be.

LESSON 66–67 | 2 × 40 MIN | ALGEBRA 8, CHAPTER III §22 | STAGE 9 · BEYOND

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Calculate the discriminant $D = b^2 - 4ac$.
- Say how many roots an equation has from the sign of D.
- Use the quadratic formula correctly, including with negative coefficients.
- Choose between factorising and the formula.

TEXTBOOK

National	Chapter III, pp. 134–140
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

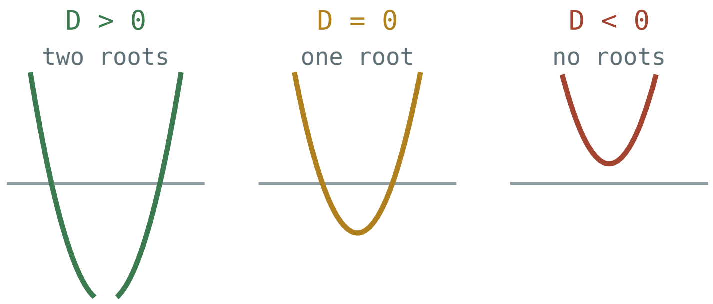
The discriminant

DEFINITION

For $ax^2 + bx + c = 0$ the **discriminant** is

$$D = b^2 - 4ac$$

D	ROOTS	THE PARABOLA
$D > 0$	two different roots	crosses the x-axis twice
$D = 0$	one root (repeated)	touches the axis
$D < 0$	no real roots	misses the axis



The sign of the discriminant decides whether the parabola crosses, touches or misses the x-axis.

Work out D **first**, every time. If it is negative you can stop — there is nothing to find, and you have saved yourself a page of arithmetic.

The formula

THE QUADRATIC FORMULA

$$x = \frac{-b \pm \sqrt{D}}{2a}, \quad \text{where } D = b^2 - 4ac$$

It comes from completing the square on the general equation, and it works for every quadratic whatever the coefficients.

THREE PLACES WHERE MARKS ARE LOST

1. $-b$ when b is already negative: for $b = -5$, $-b = +5$.
2. $-4ac$ when c is negative: $-4 \cdot 1 \cdot (-3) = +12$.
3. The whole of $-b \pm \sqrt{D}$ is divided by $2a$ — not just the root.

Which method?

- **Incomplete** ($b = 0$ or $c = 0$) → the short methods of the last lesson.
- **Factorises easily** (small whole-number roots) → factorise; it is quicker.
- **Anything else** → the formula.

A useful check: if D is a perfect square, the equation would have factorised. If it is not, the roots are irrational and the formula was the only route.

02 Worked examples

EXAMPLE 1 **Solve $2x^2 - 7x + 3 = 0$**

1. $a = 2, b = -7, c = 3$

   Read off the coefficients with their signs.

2. $D = (-7)^2 - 4 \cdot 2 \cdot 3 = 49 - 24 = 25$

   $D > 0$, so two roots.

3. $x = \dfrac{7 \pm \sqrt{25}}{4} = \dfrac{7 \pm 5}{4}$

   $-b = +7$

4. $x = 3$ or $x = 0.5$

   $\dfrac{12}{4}$ and $\dfrac{2}{4}$

ANSWER

$x = 3, x = 0.5$

EXAMPLE 2 ***Solve $x^2 + 4x + 4 = 0$***

1. $a = 1, b = 4, c = 4$

2. $D = 16 - 16 = 0$

   One repeated root.

3. $x = \dfrac{-4}{2} = -2$

4. $(x + 2)^2 = 0$

   The same answer by factorising.

ANSWER

$x = -2$

EXAMPLE 3 ***Solve $x^2 - 2x + 5 = 0$***

1. $a = 1, b = -2, c = 5$

2. $D = 4 - 20 = -16$

   $D < 0.$

3. No real roots — stop here.

   The parabola lies entirely above the x-axis.

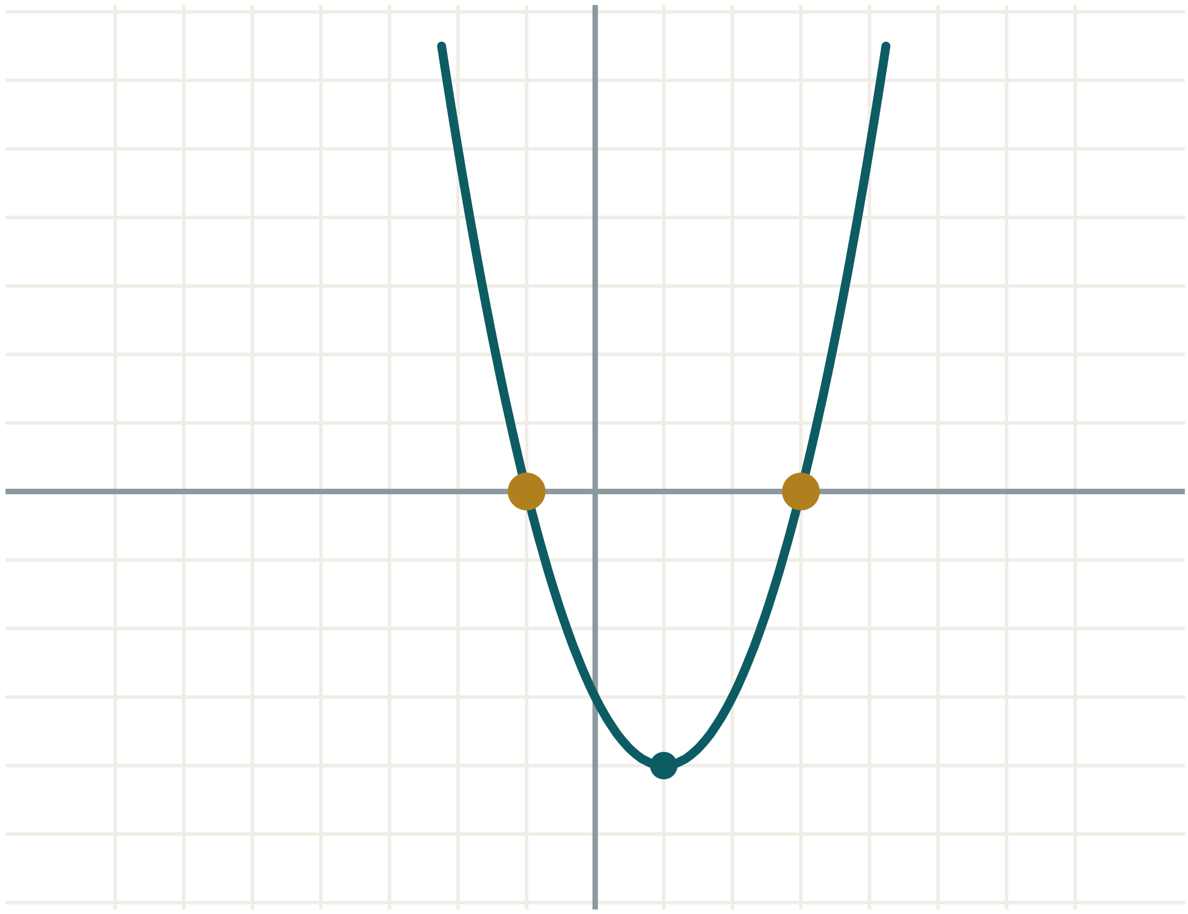
ANSWER

No real roots.

03 Interactive model

Move c downwards and watch D pass through zero as the parabola touches then crosses the axis.

D decides everything



a **1** | b **-2** | c **-3** | $D = b^2 - 4ac$ **16** | roots **two** | x_1 **-1** | x_2 **3** | $x_1 + x_2$ **2** | $x_1 \cdot x_2$ **-3**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Discriminant	Diskriminant	Дискриминант
Quadratic formula	Kvadrat tenglama formulasi	Формула корней квадратного уравнения
Two distinct roots	Ikkita har xil ildiz	Два различных корня
Repeated root	Karrali ildiz	Кратный корень
No real roots	Haqiqiy ildiz yo‘q	Нет действительных корней
Substitute	O‘rniga qo‘yish	Подставить
Simplify the surd	Ildizni soddalashtirish	Упростить корень
Exact answer	Aniq javob	Точный ответ

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

For $x^2 - 5x + 6 = 0$, D is:

- 1
- 49
- -1
- 25

$D = 0$ means:

- no roots
- one root
- two roots
- infinitely many

For $2x^2 + 3x + 5 = 0$, D is:

- 49
- -31
- 9
- 31

In the formula, when $b = -6$ the numerator starts with:

- -6
- $+6$
- 6^2
- -36

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Find D for* $x^2 + 3x + 2 = 0$

   ANSWER 1

2. *Find D for* $x^2 - 4x + 4 = 0$

   ANSWER 0

3. *Find D for* $x^2 + x + 1 = 0$

   ANSWER -3

4. *How many roots when* $D = 9$*?*

   ANSWER Two.

5. *How many roots when* $D = 0$*?*

   ANSWER One.

6. *How many roots when* $D = -5$*?*

   ANSWER None.

7. *Solve* $x^2 + 3x + 2 = 0$

   ANSWER $x = -1, x = -2$

Medium · the standard question

7 PROBLEMS

1. Solve $2x^2 - 7x + 3 = 0$

   ANSWER $x = 3, x = 0.5$

2. Solve $x^2 + 4x + 4 = 0$

   ANSWER $x = -2$

3. Solve $x^2 - 2x + 5 = 0$

   ANSWER No real roots.

4. Solve $3x^2 - 5x - 2 = 0$

   ANSWER $D = 49$: $x = 2, x = -\frac{1}{3}$

5. Solve $x^2 - 6x + 9 = 0$

   ANSWER $x = 3$

6. Solve $2x^2 + 5x - 3 = 0$

   ANSWER $D = 49$: $x = 0.5, x = -3$

7. Solve $x^2 + 2x - 8 = 0$

   ANSWER $x = 2, x = -4$

Hard · stretch and reason

7 PROBLEMS

1. *Solve* $x^2 - 4x + 1 = 0$ *exactly*

   ANSWER $D = 12$: $x = 2 \pm \sqrt{3}$

2. *Solve* $2x^2 - 4x - 1 = 0$ *exactly*

   ANSWER $D = 24$: $x = \dfrac{2 \pm \sqrt{6}}{2}$

3. *For which k does* $x^2 + kx + 9 = 0$ *have exactly one root?*

   ANSWER $k^2 = 36$, so $k = \pm 6$

4. *For which k does* $x^2 - 6x + k = 0$ *have two roots?*

   ANSWER $36 - 4k > 0$, so $k < 9$

5. *For which k does* $x^2 + 4x + k = 0$ *have no roots?*

   ANSWER $16 - 4k < 0$, so $k > 4$

6. *Solve* $5x^2 - 2x - 3 = 0$

   ANSWER $D = 64$: $x = 1, x = -0.6$

7. *Show that* $x^2 + x + 1 = 0$ *has no real roots, without the formula.*

   ANSWER $x^2 + x + 1 = (x + 0.5)^2 + 0.75 \geq 0.75 > 0$ for every x.

07 Homework

Homework — 6 problems

Algebra 8, Chapter III, pp. 134–140. Find D before anything else.

1. *Find D and solve* $x^2 - 7x + 10 = 0$
2. *Find D and solve* $3x^2 + 2x - 1 = 0$
3. *Find D and solve* $x^2 + 2x + 5 = 0$
4. *Solve* $x^2 - 8x + 16 = 0$
5. *Solve* $x^2 - 6x + 2 = 0$ *exactly*

6. *For which k does* $x^2 + kx + 4 = 0$ *have exactly one root?*